Assignment 2 Discussion
CS 772

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Define Transliteration

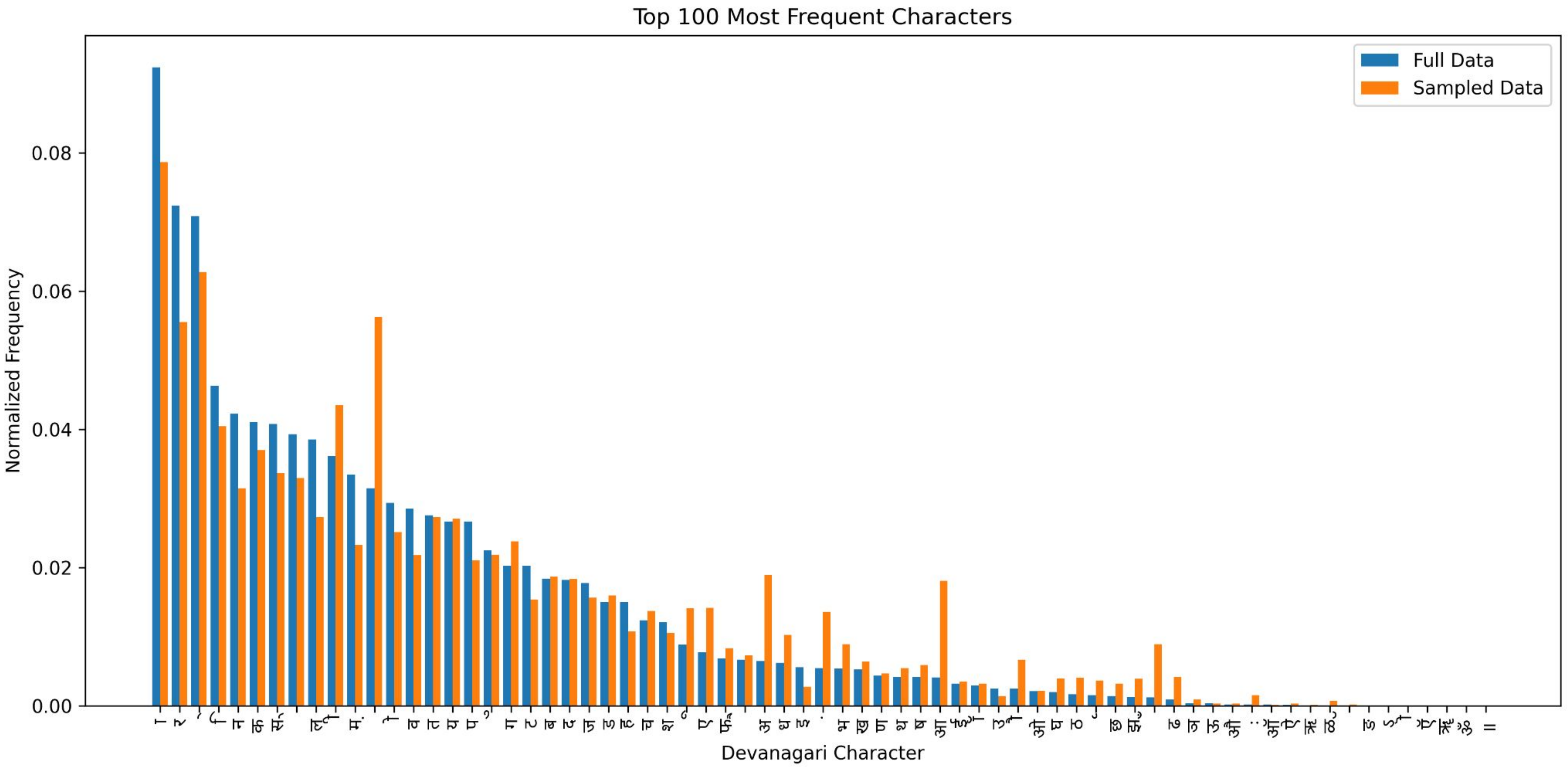
Transliteration is writing words from one language using the alphabet/script of another language, focusing on representing the sounds rather than the meaning.

- Input: Atharv aur Vartul
- Output: अथर्व और वर्तुल

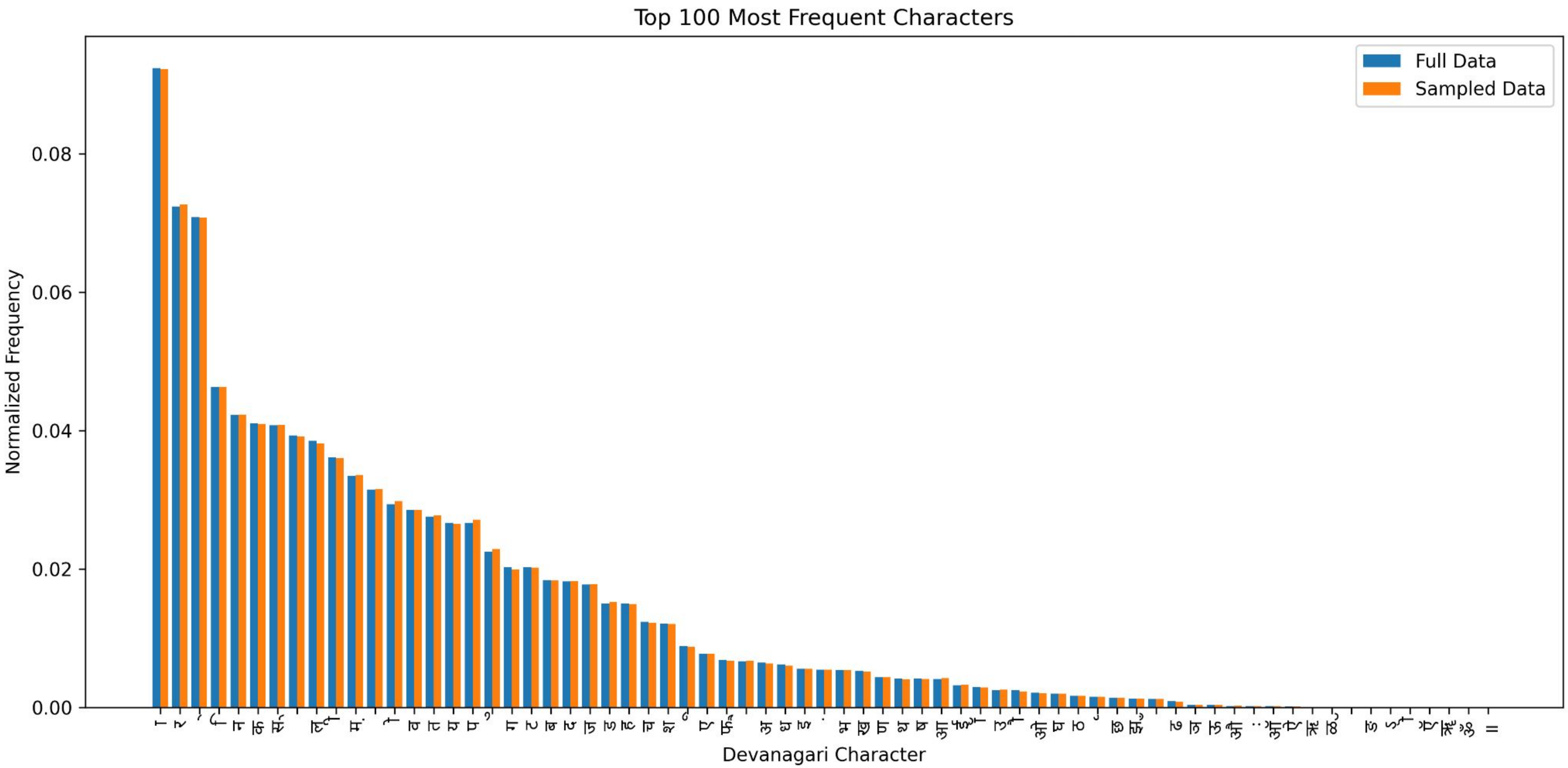
Data downloading and cleaning

- Number of training samples used = **100k**
- For sampling, earlier we tried with **hybrid sampling** method, that combines the **orthographic frequency**, **word length**, and **randomness** to select diverse examples. Words containing rarer orthographic units are prioritized, stratified by length, and supplemented with random samples to ensure balanced linguistic coverage and variety. But just using **random sampling** gave better results as the randomness brings more diversity to data.
- Sourced from Aksharantar.
- We stripped other key value pairs of json entry and used only native word and english word key.

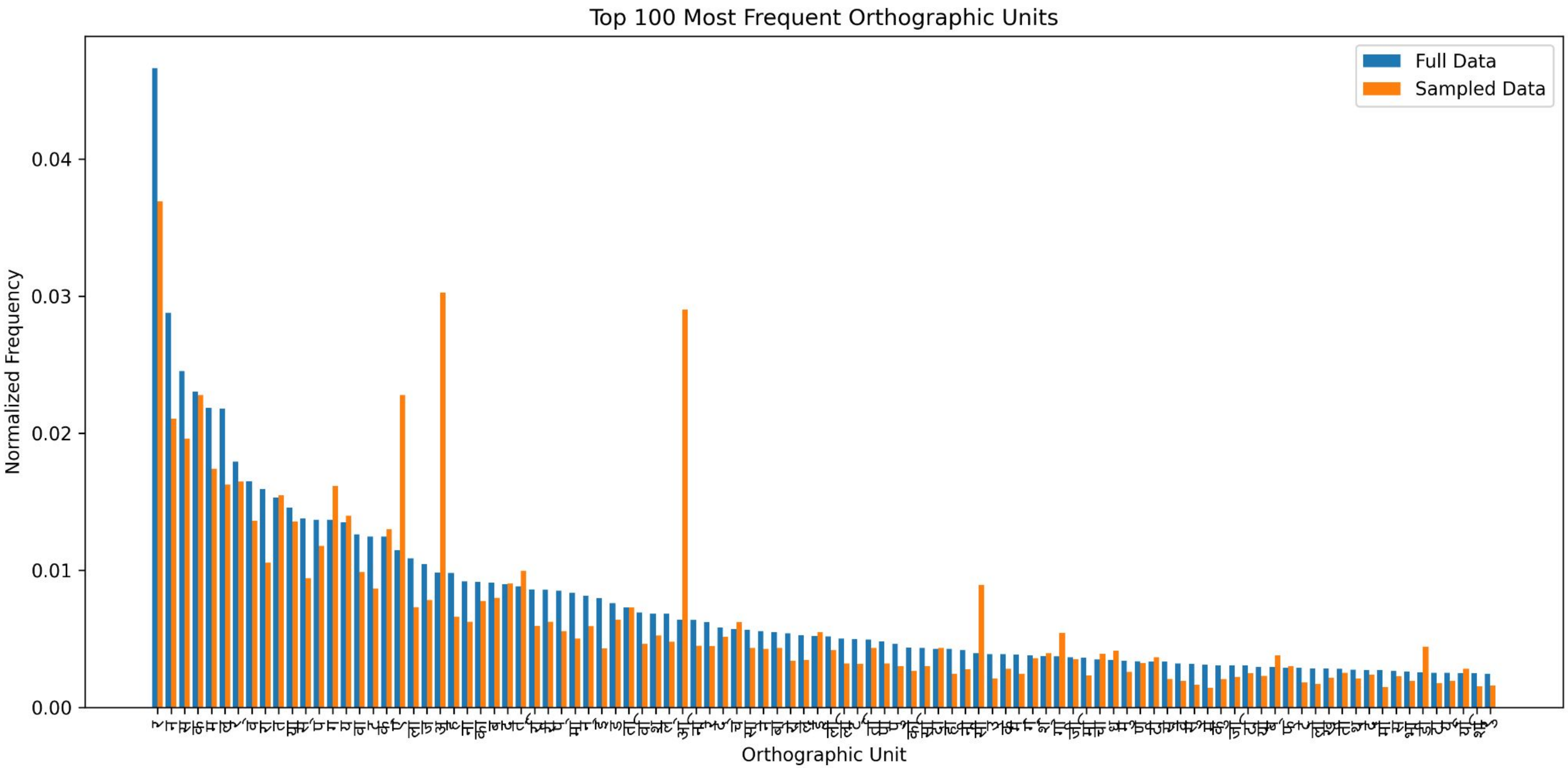
Sampling Comparison: **Hybrid** v/s Random



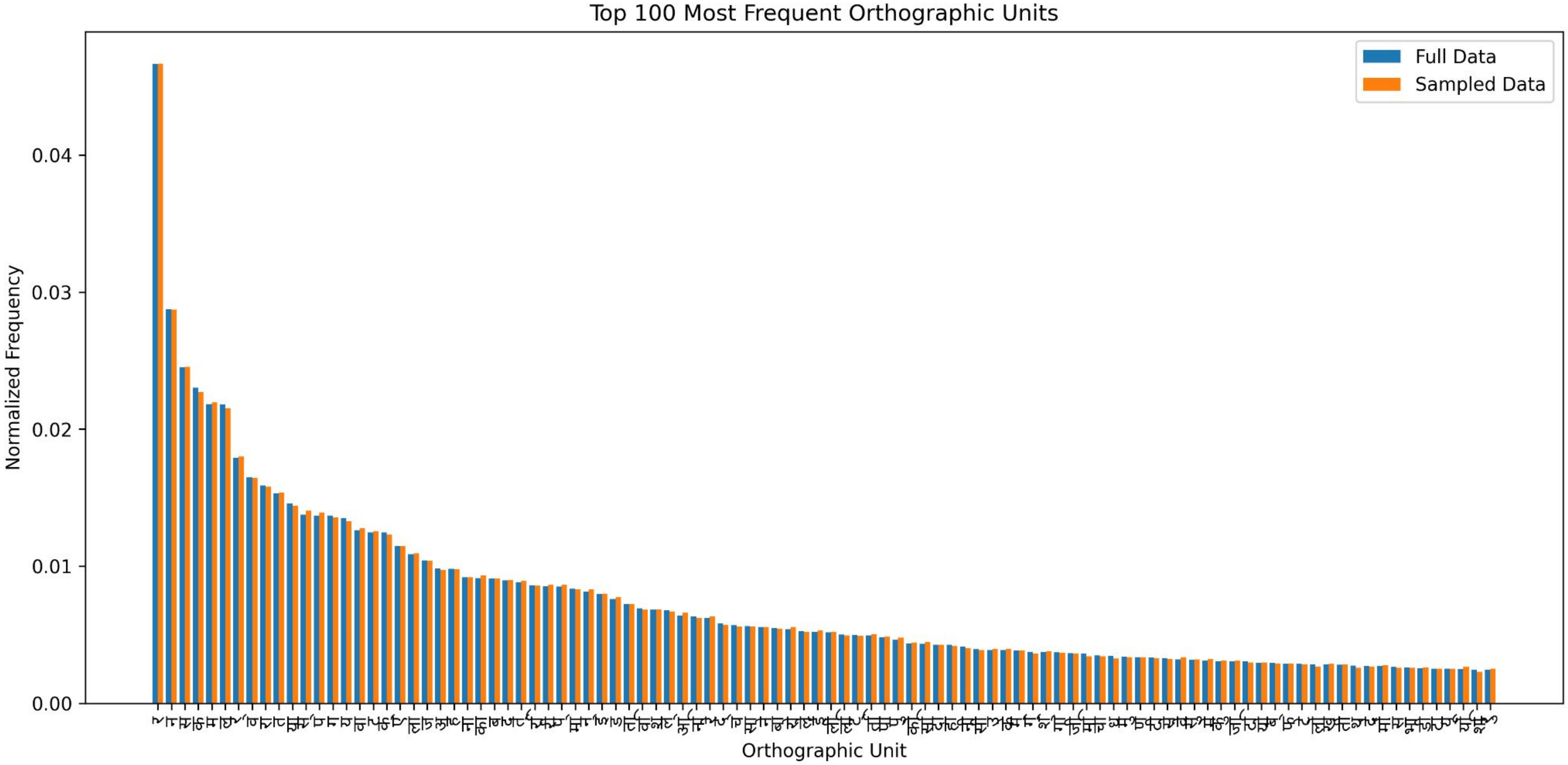
Sampling Comparison: Hybrid v/s **Random**



Sampling Comparison: **Hybrid** v/s Random



Sampling Comparison: Hybrid v/s **Random**



LSTM Based Transliteration - Hyperparameters used

Aspect	Description
Model Type	LSTM Sequence-to-Sequence
Embedding Dimension	128
Hidden Dimension	256
Number of Layers	2
Dropout	0.1
Batch Size	128
Learning Rate	0.001 (decayed on plateau)
Optimizer	Adam
Number of Epochs	50
Teacher Forcing Ratio	0.99 → 0.7 (decayed by 0.95)

https://mlarchive.com/deep-learning/understanding-long-short-term-memory-networks/

LSTM Based Transliteration

Beam Width	Word Exact Accuracy(Test)	F score(Test)
1(Greedy)	0.4627	0.8880
3	**0.4669**	**0.8889**
5	**0.4669**	**0.8889**

https://mlarchive.com/deep-learning/understanding-long-short-term-memory-networks/

LSTM Based Transliteration

- Beam search boosts the F score and accuracy marginally.
- Beam search helps a little for exact word matches, but after beam width 3, there is no significant gain.
- Increased inference time.

https://mlarchive.com/deep-learning/understanding-long-short-term-memory-networks/

Transformer Based Transliteration - Hyperparameters Used

Aspect	Description
Model Type	Transformer (Seq2Seq)
Hidden Dimension (d_model)	256
Attention Heads	8
Encoder / Decoder Layers	2 each
Feedforward Dim	512
Dropout	0.1
Batch Size	64
Learning Rate	5e-4 (decayed on plateau)
Epochs	50
Teacher Forcing	0.99 → 0.7 (decayed gradually)

https://docs.pytorch.org/docs/stable/generated/torch.nn.Transformer.html

Transformer Based Transliteration

Beam Width	Word Exact Accuracy(Test)	F score(Test)
1(Greedy)	0.4764	**0.8886**
3	**0.4782**	0.8880
5	0.4781	0.8880

https://docs.pytorch.org/docs/stable/generated/torch.nn.Transformer.html

Transformer Based Transliteration

- Beam search provides a tiny boost in word-level accuracy (~0.18%), but no improvement in F score.
- No gain after beam width 3.
- Increased inference time.

https://docs.pytorch.org/docs/stable/generated/torch.nn.Transformer.html

LLM Based Transliteration

- Model Used: **GPT-4O-mini**
- Tested with various values of temperature and top_p values.
- Zero shot prompting used.

https://platform.openai.com/docs/overview

LLM Based Transliteration - Results(Test)

Temperature	Top_p	Word Exact Accuracy	F score
0.0	0.7	0.5144	0.8929
0.0.	0.85	0.5259	0.8980
0.0	1.0	**0.5272**	**0.8984**
0.4	0.7	0.5226	0.8981
0.4	0.85	0.5232	0.8970
0.4	1.0	0.5173	0.8968
0.8	0.7	0.5175	0.8968
0.8	0.85	0.5116	0.8951
0.8	1.0	0.4994	0.8882
1.2	0.7	0.5195	0.8966
1.2	0.85	0.5035	0.8916
1.2	1.0	0.4533	0.8642

LLM Based Transliteration

- Lower temperature (0–0.4) and Top-p ≈ 0.85–1.0 gives the best transliteration results.
- High temperature hurts exact match accuracy and F-score.

Error Analysis

1. Getting confused between sound of (त थ द ध न) and (ट ठ ड ढ ण)
   (त थ द ध न) is pronounced with tongue touches teeth (dental)
   (ट ठ ड ढ ण) is pronounced with tongue curled back (retroflex)
   example
      tamatar : तमातर (LSTM), तमातर (transformer)
      damru: दामरू (LSTM), दमरू (Transformer)
      dabba : दब्बा (LSTM), दब्बा (Transformer)
      dhakkan : धक्कन (Transformer)
      theek: थीक(LSTM)
      thakaan: ठकां (Transformer)
      ganana: गनाना (LSTM), गनाना (Transformer)

Error Analysis

2. getting confused between अ and आ.
    a. kaam kam : काम काम (LSTM)
    b. van : वान (Transformer)
    c. arvind : आरविंद (Transformer)
    d. bhaarat bharat : भारत भरत (Transformer) , भारत भारत (LSTM)
3. treating small words as abbreviation, example
    e. to : टीओ (LSTM)
4. Hallucination:
    f. ko: कोब (LSTM)

Project Proposal

Problem Statement: Investigate methods to convert a pre-trained small LM into an MoE model, train a lightweight gating mechanism, and measure trade-offs between efficiency, accuracy, and interpretability.

Motivation: LLM models consists of dense MLP layers that are heavy weighted. During inference not all of these weights give valuable contribution for next token generation, but since whole model has to be loaded, it requires a lot of GPU usage. Introducing MoE can help to activate only those Experts that contributes most to output for the next token.

Methodology: For this project we will go with a basic causal language modeling task. We plan to compare different MoE variants (like upcycling, virtual group, Drop-Upcycling) vs. dense models (gemma 270M) on inference latency, model size (parameters, memory footprint), and accuracy (or any baseline depending upon task) while measuring routing/load-balance behavior.

Paper referenced: Upcycling Large Language Models into Mixture of Experts
Drop-Upcycling: Training Sparse Mixture of Experts with Partial Re-initialization
DeRS: Towards Extremely Efficient Upcycled Mixture-of-Experts Models

Home | Sign out |
Academic
Canara Bank Authorisation Letter
Alumni form
Pay Convo Fees
Convo Form
Change of tag
All About Courses
Account/Fee
Classroom Attendance
Course Feedback
Guidelines
Registration
Course Prerequisites
Academic/Personal Details
View
Update/Apply for scholarship
Vote for Excellence-in-Teaching
Faculty Advisor
Fees Status/Scholarship
TimeTable
Library/Archives
Placement
SAP user license details
ERP Master data
Accounts and Salary
Moodle
IRCC
Surveys
Admin Interfaces
Maintenance complaints
Visitor Entry/Exit Pass
GatePass
Guest House
Downloadable Forms
Online Purchase System (Reports Only)
Scrutiny of job applicants
Online Mobile Registry
Print

Account/Fee
Classroom Attendance
Course Feedback
Guidelines
Registration
Course Prerequisites
Academic/Personal Details
View
Update/Apply for scholarship
Vote for Excellence-in-Teaching
Faculty Advisor
Fees Status/Scholarship
TimeTable
Library/Archives
Placement
SAP user license details
ERP Master data
Accounts and Salary
Moodle
IRCC
Surveys
Admin Interfaces
Maintenance complaints
Visitor Entry/Exit Pass
GatePass
Guest House
Downloadable Forms
Online Purchase System (Reports Only)
Scrutiny of job applicants
Online Mobile Registry
Print
Account/Fee
Classroom Attendance
Course Feedback
Guidelines
Registration
Course Prerequisites
Academic/Personal Details
View
Update/Apply for scholarship
Account/Fee
Classroom Attendance
Course Feedback
Guidelines
Registration
Course Prerequisites
Academic/Personal Details
View
Update/Apply for scholarship
Vote for Excellence-in-Teaching
Faculty Advisor
Fees Status/Scholarship
TimeTable
Library/Archives
Placement
SAP user license details
ERP Master data
Accounts and Salary
Moodle
IRCC
Surveys
Admin Interfaces
Maintenance complaints
Visitor Entry/Exit Pass
GatePass
Guest House
Downloadable Forms

Roll no	24M0834	**Name**	Shingewar Atharv Hemant Snita
Department	Computer Science and Engineering	**Program**	M.Tech.
Status		Active	
Program Category	Research Assistant - Research Assistantship		
Program validity date	2027-07-15	**Semester validity date?**	2027-01-31
Faculty Advisor(a)	Shivaram Kalyanakrishnan		

Table of Contents

Academic Performance Summary

Year	Sem	SPI	CPI	Sem Credits Used for SPI	Completed Semester Credits	Cumulative Credits Used for CPI	Completed Cumulative Credits
2025	Spring	10.0	9.15	16.0	16.0	54.0	54.0
2025	Autumn	10.0	8.79	6.0	6.0	38.0	38.0
2024	Spring	9.0	8.56	12.0	12.0	32.0	32.0
2024	Autumn	8.3	8.3	20.0	20.0	20.0	20.0

Semesterwise Details

*This registration is subject to approval(s) from faculty advisor/Course Instructor/Academic office.

Year/Semester: 2025-26/Spring

Course Code	Course Name	Tag	Credits	Grade	Crdit/Audit
CS 692	R & D Project II	Department elective	6.0	AA	C
CS 694	Seminar	Core course	4.0	AA	C
CS 728	Organization of Web Information	Department elective	6.0	AA	C
CS 899	Communication Skills	Core course	6.0	PP	N

Year/Semester: 2025-26/Autumn

Course Code	Course Name	Tag	Credits	Grade	Cr
CS 601	Algorithms and Complexity	Additional Learning	6.0	BB	C
CS 691	R & D Project	Department elective	6.0	AA	C
CS 772	Deep Learning for Natural Language Processing	Additional Learning	6.0	AB	C

Year/Semester: 2024-25/Spring

Course Code	Course Name	Tag	Credits	Grade	Cr
CS 695	Topics in Virtualization and Cloud Computing	Department elective	6.0	AB	C
CS 736	Medical Image Computing	Department elective	6.0	AB	C

Year/Semester: 2024-25/Autumn

Course Code	Course Name	Tag	Credits	Grade	Cr
CS 699	Software Lab.	Core course	8.0	BB	C
CS 725	Foundations of Machine Learning	Department elective	6.0	AB	C
CS 744	Design and Engineering of Computing Systems	Department elective	6.0	BB	C
GC 101	Gender in the workplace	Core course	0.0	PP	N
TA 101	Teaching Assistant Skill Enhancement & Training (TASET)	Core course	0.0	PP	N

Personal Info

Atharv Hemant Shingewar Rollno : **24M0834**	Room : H17-3004
Mobile :9987708019	
M.Tech.,Computer Science and Engineering	
DoB : 2000-05-08	Nationality: Indian

Informtion last updated on 2024-07-24

Parent/Guardian Details	Father : HEMANT SHINGEWAR, 9082770400, omfertilizers2014@gmail.com
Present Home Address	B-205, SUNRISE GALAXY CHS SANTOSHI MATA ROAD, RAMBAUG - 4, NEAR VIJAY SALES Kalyan MAHARASHTRA 421301
Emergency Contact	HEMANT SHINGEWAR, 9082770400 B-205, SUNRISE GALAXY CHS SANTOSHI MATA ROAD, RAMBAUG-4, KALYAN(W) 421301
Local Guardian	HEMANT SHINGEWAR, 9082770400 B-205, SUNRISE GALAXY CHS SANTOSHI MATA ROAD, RAMBAUG - 4, NEAR VIJAY SALES MAHARASHTRA Kalyan 421301

Student generated forms

1. Student Declaration Form
2. Hostel Registration Form
3. Medical File Form + Immunisation record